

What is training on dep-0?

One CAT generalist is being fine-tuned for leg, waist and arm control across original CAT scenes and dense clutter.

LIVE AT SNAPSHOT: 1,138,753,536 transitions · 8,192 environments · one W&B run

Released CAT generalist

Native pretrained actor + critic.
Leg/navigation weights restored;
new upper-body outputs added.

One trainable whole-body actor

406 observations → 29 actions.
12 leg + 3 waist + 14 arm/wrist.
Dex3 fingers stay fixed.

CAT simulator + native PPO

39 fixed scenes in one learner.
50 Hz control, 500 Hz physics.
Continuous updates until stopped.

On-policy rollouts + PPO gradients

Implemented additions

Arm and waist actuation; mesh-sized hand envelopes including thumbs; predictive hand-clearance features; hand/arm penalties; dense tables/chairs and generic clutter.

All run together with the original CAT reward and scene sampler.

What the results currently support

Generic clutter survival improves. Furniture remains difficult:
97.42% of its recent completed episodes terminate early.

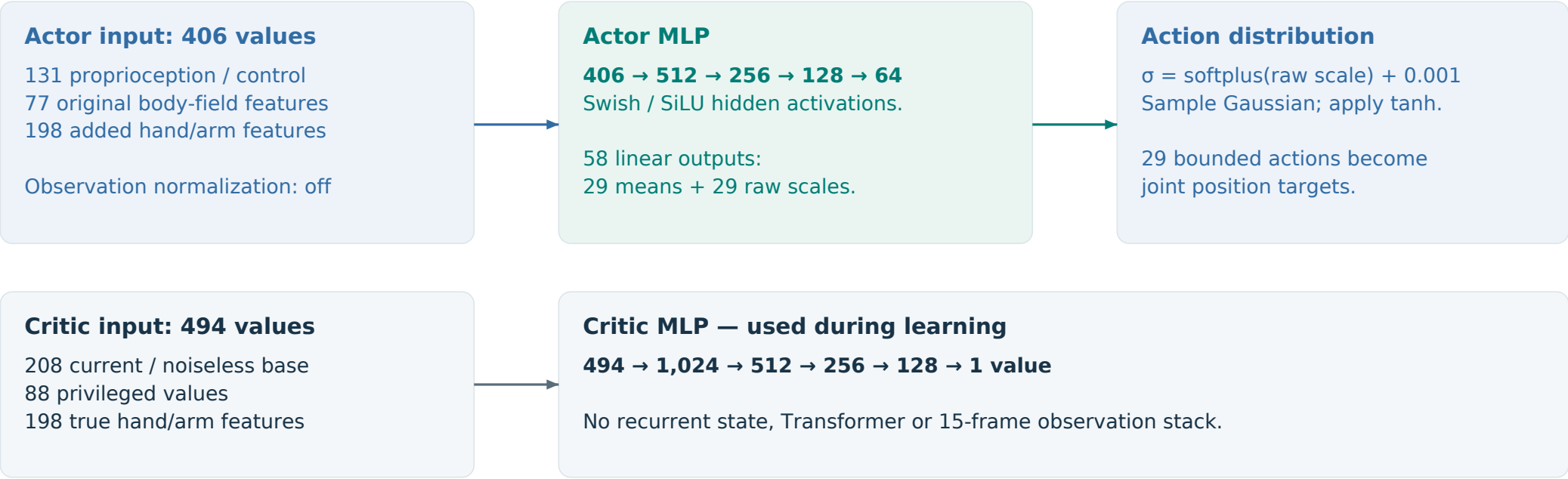
No human motion demonstrations or imitation loss are used.
Reliable protective hand raising is not yet demonstrated.

[Open the single live W&B run](#)

Snapshot: 15 Sep 2026, 14:47 UTC / 18:47 Dubai

The actor sees fields and controls 29 joints

Separate feed-forward actor and critic MLPs preserve CAT’s hidden-layer sizes. Every layer remains trainable.

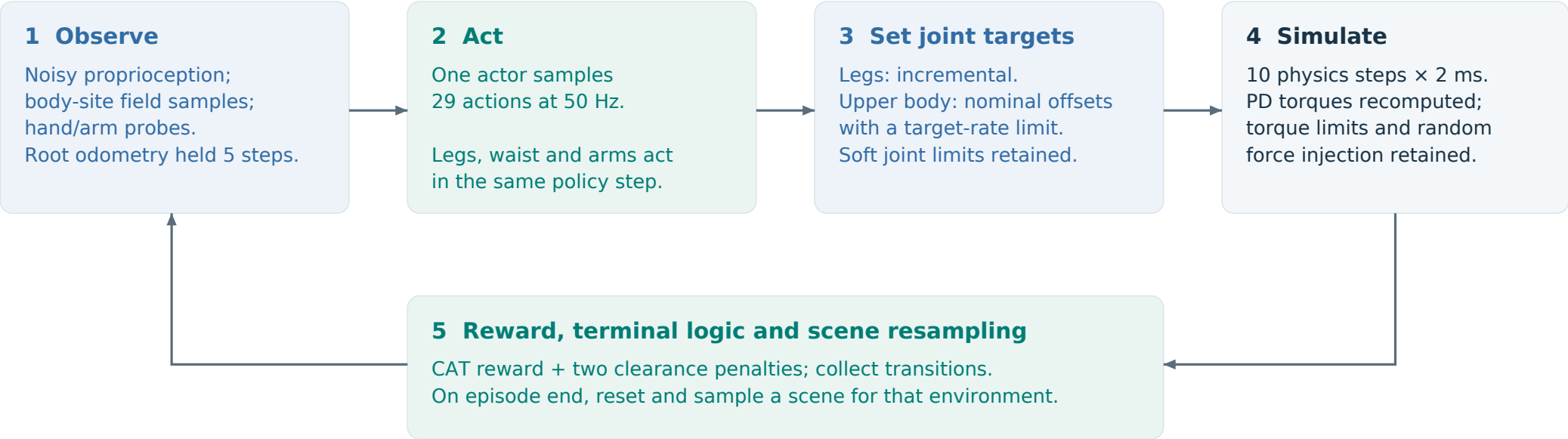


Feature block	Where it comes from	Purpose
11 original sites × 7 = 77	Head, torso, pelvis, limbs	Guidance xyz + boundary xyz + SDF
22 new probes × 9 = 198	Hand envelopes, palms, arms	3 distances + boundary xyz + 3 metadata

Warm start: released inputs are 162 actor / 250 critic values, with 12 leg actions. Retained weights are copied; new input weights and action means start at zero, new action standard deviation at 0.05. Mapped retained actor/value outputs match at initialization; physical trajectories need not match. New arm skills must be learned.

From field queries to motion: every 20 ms

The policy observes a known static 3D field map. This training run has no camera, depth reconstruction or online SLAM.

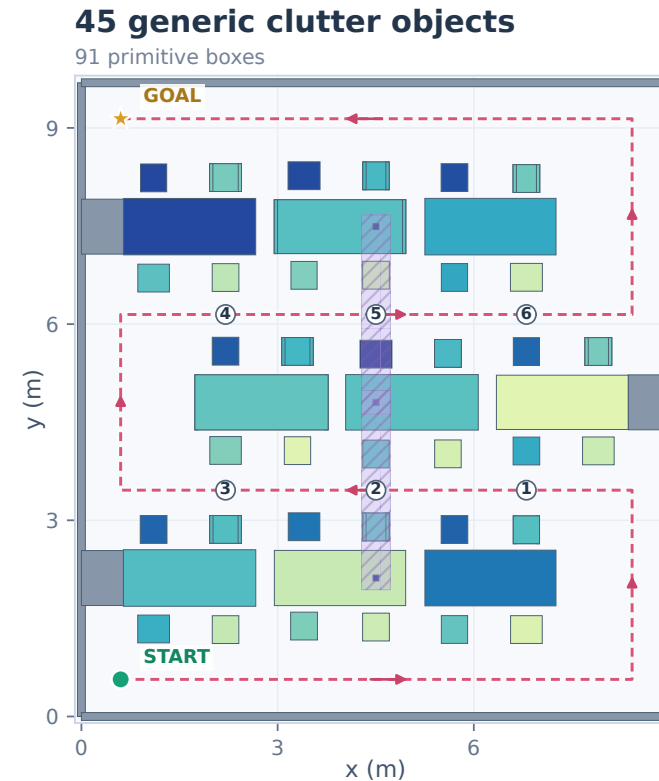
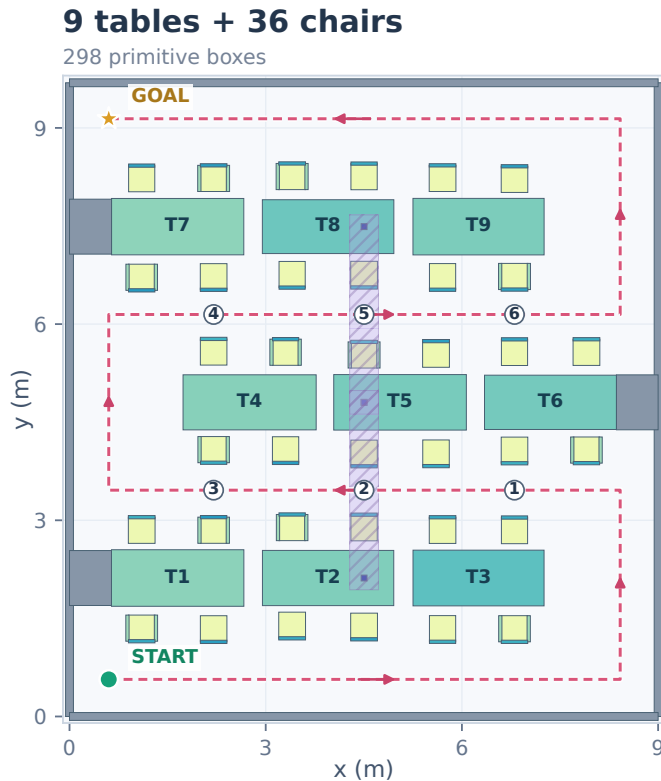


Action group	Target update	What moves
12 leg joints	$q_{\text{target}} = q_{\text{previous}} + 0.5 \times \text{action}$	Original CAT incremental semantics
17 upper-body joints	$q_{\text{target}} = q_{\text{nominal}} + 0.8 \times \text{action}$	3 waist + 14 arm / wrist joints
Upper target rate	2 rad/s = at most 0.04 rad/control step	Smooths target changes before limits
Dex3-1 fingers	Fixed pose; 7 finger joints welded per hand	3-finger hardware; no finger actions

Timing detail: root odometry refreshes every 0.1 s; articulation and field queries update each control step. The critic uses the current true state. No active odometry-noise injection is applied in this path.

The added rooms require repeated narrow passages

Actual current bank geometry: each room is 9.00×9.69 m, with six 68–78 cm bottlenecks and obstacles at several heights.



Primitive top height (m)

1.4
1.2
1.0
0.8
0.6
0.4

9.00 × 9.69 m
4 cm voxels

Dashed pink: geometric reference route • Hatched purple: overhead obstruction • Circled numbers: six narrow passages

One fixed bank: 39 scenes

36 byte-verified public CAT fields
+ 1 reconstructed missing CAT slot
+ 1 furniture + 1 generic room.
Resets randomize state, not layout.

Original CAT field pipeline

3D occupancy at 4 cm resolution
→ signed distance field (SDF)
→ boundary gradient + progressive
3D fast-marching guidance to goal.

Geometry is not a learned path

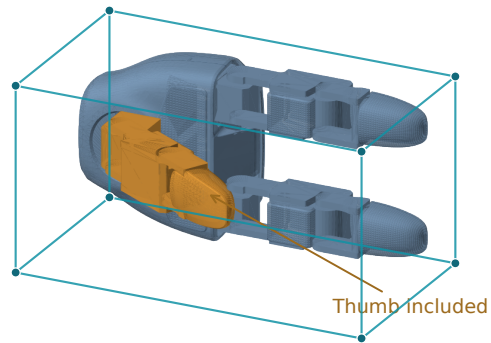
The dashed 39.85 m route checks
root clearance during construction.
It is not supplied to the actor and
does not prove dynamic feasibility.

Hand protection uses geometry and predicted clearance

Dex3-1 envelopes include the thumb and 5 mm padding. Twenty-two probes cover both hands and both arms.

Fixed Dex3 hand and safety envelope

Native STL geometry • fingers unactuated



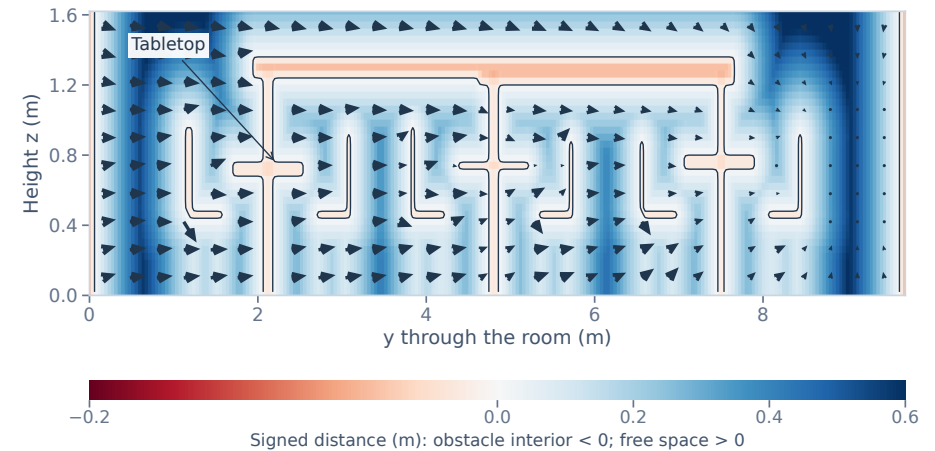
5 mm padding on every side • all 8 corner probes shown

Probe placement: each hand has a 3 cm palm sphere plus 8 box-corner points. Four 5 cm spheres cover forearms/elbows: 18 hand + 4 arm probes total.

At each probe: clearance now, at 0.2 s and 0.4 s; boundary xyz; observation age; unknown flag; position uncertainty. The last two are zero in this run.

Actual 3D CAT field through the furniture

Vertical slice at $x = 4.48$ m; arrows show projected guidance



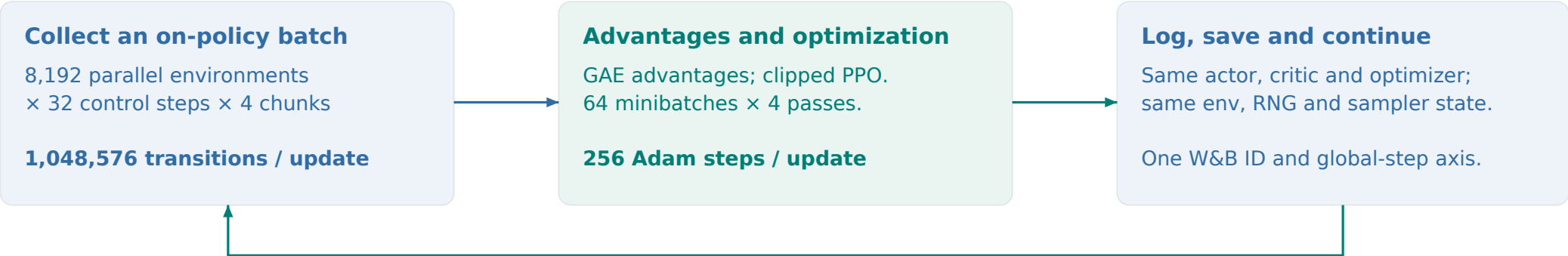
Hand penalty: below 12 cm minimum predicted clearance, multiply squared deficit by $(1 + \text{urgency})$, with urgency clipped to 0–10; average over 18 probes, scale -5 .

Arm penalty: below 8 cm current clearance, average squared deficit over 4 probes, scale -2 . Both join the full CAT reward before its 0.02 s scaling and clipping.

What this can teach: the actor can raise, tuck or reposition arms when that improves return. Prediction extrapolates current velocity; it does not simulate future actions. Point probes do not certify a whole swept volume. There is no explicit human-style hand-raising target or hard action safety filter.

Native CAT PPO runs continuously on one GPU

The released final generalist has already completed distillation. This run starts direct PPO fine-tuning; there is no live teacher stage.



Setting	Released CAT	Active dep-0 run
Parallel environments	65,536	8,192
Trajectories per minibatch	2,048	512 (16,384 transitions)
Transitions per PPO update	4,194,304	1,048,576
Unroll / minibatches / passes	32 / 64 / 4	32 / 64 / 4
Learning rate / entropy / discount	0.0003 / 0.003 / 0.98	Same
Clip / GAE / max gradient norm	0.2 / 0.95 / 1	Same
Reward scaling / advantage normalization	1 / enabled	Same

Resource change: the batch is four times smaller to fit expanded observations on the 32 GB GPU. Adam initializes once at fine-tuning start. There is no finite step budget, automatic evaluation, stage reset or rollback to best weights.

Preserved CAT task, with explicit extensions

The full CAT reward, original reset behavior, noise, disturbances, gait logic and adaptive mixed-scene wrapper remain in use.

Reward: 22 original terms + 2 additions

Original navigation / orientation and body-motion terms; gait/contact/clearance/slip/balance terms; smoothness, joint-limit and torque costs; body guidance and distance-field terms.

Task reward = clip($0.02 \times \text{weighted sum}$, 0, 10,000).

The new hand and arm costs are inside that sum. CAT's outer wrapper zeros learner reward on terminal transitions.

When an environment resets

Falls and invalid state terminate immediately. Selected self-contacts, body-field penetration and added probe penetration terminate after the original grace period.

Collision threshold: 0.0 m.

Grace: 50 control steps (about 1 s).

Timeouts are truncations; CAT's native GAE handling remains.

Reset and episode length

Original scenes: CAT start distribution (± 1 m XY), original randomization, 1,000-step / 20 s time limit.

New rooms: entrance reset with ± 8 cm XY jitter, original yaw/joint/velocity randomization, 4,000-step / 80 s limit; goal reward uses the room's actual goal.

CAT adaptive scene sampling

At each step, update per-scene completed and timeout counts with EMA decay 0.95.

$s = \text{timeout EMA} / \text{completed EMA}$

$\text{weight} = \max(1 - s, 0.001)$; normalize over 39 slots.

A reset samples from these weights. Initial weights are uniform; scene mix evolves with survival.

Critical interpretation: reaching the goal does not itself end an episode. The sampler's "success" statistic means timeout survival. Training still uses CAT's flat-floor/feet contact physics; furniture obstacles are fields, not contact-force bodies.

Runtime, unified logging and checkpoint handling

One detached process on dep-0 uses a frozen source checkout. Reporting here is read-only and does not interrupt training.

Hardware / software	Actual setup
Host / process	dep-0 (WS012743) / PID 11525
GPU	RTX 5090 · 32,607 MiB total
NVIDIA allocated VRAM	30,425 MiB = 29.71 GiB
JAX peak live / pool limit	22.64 / 28.26 GiB
Epoch throughput	≈87,600 transitions/s
CPU / host RAM	24 logical CPUs / 125 GiB
Python / JAX / Brax	3.12.9 / 0.4.38 / 0.12.3
MuJoCo + MJX / Flax	3.3.1 / 0.10.4

VRAM accounting overlaps: the CUDA total includes the allocator pool. Do not add these values. Execution samples reach 99–100% utilization; host checkpoint writes create gaps. The snapshot caught 0% at a boundary. Throughput excludes checkpoint-writing time.

One W&B experiment

Run ID **1d39c55c** / project CAT-wholebody.
Pooled episode and optimization metrics plus per-scene diagnostics share one global-step axis. Scene families are not separate runs.

One selected best model

checkpoints/best contains actor/critic inference state.
Selected by maximum single-update training reward proxy.
Current best: update 6 / 6,291,456 transitions, score 0.350549. It never replaces the live learner.
This is not a held-out furniture-success score.

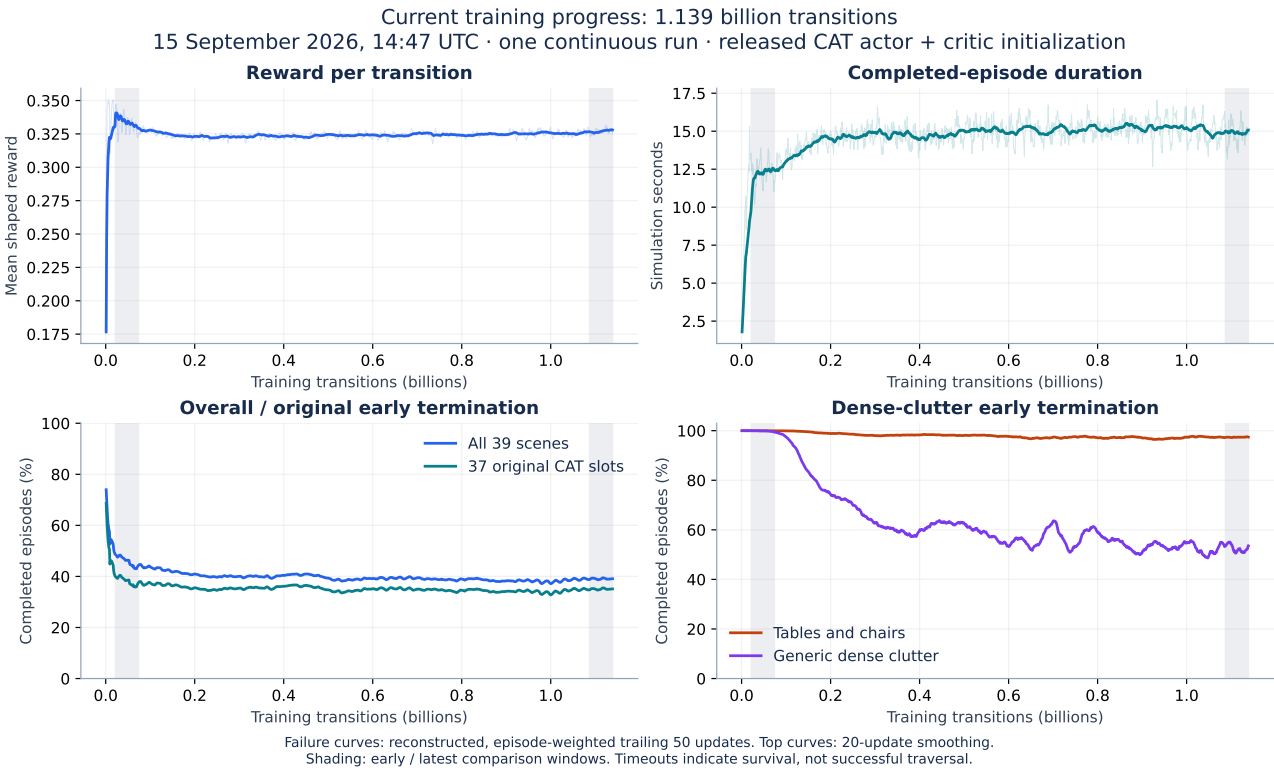
One overwritten full recovery file

resume.msgpack stores current weights, Adam, RNG, normalizer, environment, scene sampler and metric buffers.
Atomic save at initialization and after each PPO update.
About 1.03 GiB; native best model about 6 MB.
Exact resume checks source, fields and configuration.

Run: outputs/cat_wholebody_generalist_20260915. **Start:** 10:37:15 UTC, 15 September. Same process, attempt 1, zero logged nonfinite events through the snapshot. Continuous means no scheduled cutoff; process errors can still stop it.

Measured progress: survival improves, furniture lags

Read-only snapshot at 14:47:33 UTC: 1,086 complete PPO updates and 1.556 million completed training episodes.



Recent comparison	Early	Latest
All: early termination	44.30%	39.10%
Original CAT slots	37.54%	35.10%
Furniture	99.96%	97.42%
Generic clutter	99.51%	53.54%
Reward / transition	0.3322	0.3270
Logged duration (s)	12.30	14.85

Hand objective: proximity-penalty magnitude per logged episode step is 26.38% higher. That is worse, despite longer episodes. Reward per transition is 1.59% lower.

What the comparison measures

Early = updates 21-70; latest = 1037-1086 (50 updates each). Early is already fine-tuned, not the pretrained baseline. Failure combines falls, collision and invalid state. Counts are episode-weighted after undoing the logger’s rolling means.

What it does not establish

Adaptive training scene mix is not a matched evaluation. Timeout is not goal success. Neither safe hand raising nor reliable furniture traversal is demonstrated.

What remains to establish, and how to reproduce it

The implementation is inspectable and the run is reproducible. Learned skill quality still requires controlled evaluation.

Evidence already obtained

243 full regression checks and 72 later focused recovery checks passed. Tests cover original CAT reset/step behavior, exact mapped actor/critic outputs, fields, sampler parity and mixed-scene reset/step integration.

The actual 8,192-environment GPU profile compiled and has executed 1,086 updates. These are integration checks, not traversal-success tests.

Still missing from the experiment

A matched pretrained-versus-current evaluation; goal completion and hand-clearance statistics; unseen room layouts; collision-force evaluation with physical furniture; a dynamic whole-body feasibility check.

No human motion data or imitation loss is integrated. Scene-aligned, retargeted human motion remains a proposed follow-up.

SOURCE AND DATA IDENTITIES

Component	Pinned identity
Training source	40b4dd2682dd4e5d75a6b853c24333da99379cf3
Original CAT reference	866ba392f1c1e84b92ad75fa66550f26e8af8e48
Pretrained model	46ce4b57ba0639168d51741b661ff62f7ce6f045 · checkpoint 005033164800
Released field dataset	db8c202fa724bc4d3dba2cb9fead1267f15fd811
Missing original scene	D8G2L3O2S13 reconstructed; unknown original bytes are not claimed

Frozen training entry point

Whole-body task implementation

Released pretrained checkpoint

The accompanying Markdown has the exact launch command, operator paths, field/source hashes and software lock. Figure inputs, geometry provenance, validated progress counts and the PDF builder are saved with this report. All plots show the current fields or measured training logs.